

Bridging Embodied AI and Accessibility

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Abstract

Embodied AI systems are creating new ways to support people with diverse accessibility needs in physical assistance, communication, navigation, rehabilitation, caregiving, and daily life. However, progress in these areas depends on more than better robot autonomy, sensing, or control. It also requires new interaction techniques and systems that shape how people and intelligent technologies work together: how users express goals, how robots provide assistance, how control is shared, how systems adapt to different abilities and contexts, and how failures are detected, explained, and repaired. This workshop brings together researchers across user interfaces, machine learning, robotics, assistive technology, accessibility, rehabilitation engineering, AR/VR, wearables, and participatory design to examine the intersection of assistive embodied AI systems and accessibility. Topics include physical assistance and caregiving, robot teleoperation, shared autonomy, speech and multimodal control, AI-supported visual access and navigation, and disability-led methods for real-world deployment. Through paper presentations, lightning talks, demos, breakout discussions, and synthesis activities, the workshop will identify open challenges, design principles, and research opportunities for building assistive embodied AI systems that are usable, trustworthy, adaptable, and centered on user agency in real-world contexts.

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CCS Concepts

• **Computer systems organization** → **Robotics**; *Embedded and cyber-physical systems*; • **Human-centered computing** → **Accessibility**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

Embodied AI, embodied interaction, multimodal interaction, assistive technology, assistive robotics, human–robot interaction, intelligent user interfaces, cyber–physical systems, human augmentation

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1 Workshop Motivation and Scope

1.1 Physical Assistance, Caregiving, and Activities of Daily Living

Assistive robots can support people with mobility or upper-limb impairments in activities of daily living—eating, drinking, object retrieval, and other manipulation tasks often requiring caregiver assistance [6, 29]. Robot-assisted feeding illustrates how success depends on both reliable acquisition and safe transfer [12]. These systems combine manipulation, physical human–robot interaction, perception, and tactile sensing for domestic environments, yet technical capability alone does not ensure usefulness or acceptance. Ethnographic work on robotic eating/drinking assistance highlights structural, social, and interaction requirements—safety, privacy, ease of use, and independence from continuous caregiving [33]. Systems should enhance independence while adapting to users’ abilities, routines, environments, and preferred forms of assistance.

1.2 Accessible Robot Control, Teleoperation, and Shared Autonomy

Accessible robot control requires interfaces that accommodate heterogeneous motor abilities while preserving agency. Conventional teleoperation often demands multi-degree-of-freedom input or mode switching that is difficult for people with upper-limb impairments. Body-machine interfaces provide control from remaining movement combined with shared autonomy [18]; customizable interfaces [40, 41] and shared-control methods [1, 11] reduce input complexity by combining low-dimensional commands with context-dependent assistance. Human-in-the-loop adaptation [14]

and learned low-dimensional action spaces [22] further map accessible inputs onto high-dimensional motions. Adaptive DoF mapping lets users select, reject, or override suggestions while controlling many degrees of freedom through one- or two-dimensional inputs [34, 38]. Effective intent communication extends beyond trajectories to robot state, attention, and instructions tied to the robot, environment, or user [35]. Evaluations with joysticks, assistive buttons, and head-based input show that modality, feedback, and autonomy must be individually tailored [31, 39], while involving users, caregivers, and clinicians helps ensure shared autonomy supports performance without making users passive supervisors [13].

1.3 Speech, Language, and Multimodal Interfaces for Embodied AI

Interaction proxies translate user intent across physical, touch-screen, and immersive environments [20, 21], remapping input to users’ abilities, contexts, and preferred modalities such as speech [21]. LLM-supported speech systems, including [8, 9, 32, 55], enable hands-free tasks through context tracking. Related work develops speech and multimodal interfaces for BLV users and wheelchair users [25, 26, 28], combines gaze and speech [19], surveys modalities for senior care [42], and examines wheelchair interfaces spanning speech, gesture, and brain commands [44]. Affective touch [30] and multimodal mouth-gesture recognition for feeding bedridden users [7] further expand embodied control. Nonverbal communication is also crucial: socially assistive robots can support people with speech and motor disabilities through movement, gesture, and spatial positioning [52], while shared object awareness can enhance social interaction [16, 43]. Further research should examine how embodied systems coordinate modalities, adapt to users, and make physical interaction legible, accessible, and socially meaningful.

1.4 AI Perception, Visual Access, and Navigation for Blind and Low-Vision People

Although AI systems have advanced in visual and spatial understanding, their capabilities do not always match BLV users’ real-time needs. Navigation UIs for collision avoidance [27], vision-language systems that describe surroundings and support everyday tasks [15], and context-aware models that use user actions and embodied context to prioritize information [3–5] point toward continuous assistive support with implications for personal AI, ridesharing, navigation, and autonomous mobility [10]. Scene descriptions alone may not provide actionable information in the moment, requiring advances in embodied perception, human–AI interaction, and accessibility for dynamic environments.

1.5 Augmented, Virtual, and Mixed Reality Interfaces for Embodied Interaction

AR/VR/MR can provide spatially registered interfaces for robot control, motion and perception communication, and embodied prototyping without continuous physical robot access. Virtual design elements represent robot state, actions, tasks, environments, and interaction possibilities [54]; virtual bodies and augmented overlays communicate directions, mappings, and obstacles in the workspace [36, 37]. AR supports teleoperators in manipulation [2]

and can present intended motions in the shared environment [53], while spatial registration helps users connect intent information to physical action [35]. Transitional XR environments such as Adaptive support, shared control, feedback, and alternative inputs across virtual, mixed, and physical settings [34]. These systems enable low-dimensional, head-based, or switch-based control for users who cannot use standard multi-degree-of-freedom input; hands-free facial-expression controllers co-designed with people with motor impairments demonstrate such modalities in real-time interaction [49, 50]. XR simulation can support early user involvement and reduce logistical and safety barriers, but immersive embodiment is not inherently accessible: customized feedback shapes whether unfamiliar virtual abilities feel comfortable and meaningful [45].

1.6 Disability-Led Design, Agency, Ethics, and Real-World Evaluation

Assistive robotics increasingly advocates for disability-centered design that engages disabled people as collaborators throughout development and evaluation [17, 23, 24, 52] and challenges paternalistic approaches that prioritize safety or efficiency over autonomy [13]. Autoethnographic and lived-experience work shows how personal need and embodied knowledge drive design while surfacing agency and emotional labor [48], and evaluation must extend beyond efficiency to sensation, physical agency, and felt experience [47]. As embodied AI mediates communication, AI-generated language can reduce AAC effort while raising concerns about voice and intent [51], and AI-mediated guidance shapes autonomy and social bias recognition [46]. These threads motivate personalization, trust, safety, and deployment in homes, clinics, and communities.

2 Workshop Program Format and Schedule

We propose a full-day workshop for approximately 25–50 participants. The program combines paper sessions, a demo/poster session, a panel, thematic breakouts, and collaboration pitches to surface shared research challenges and foster cross-community collaboration. Participants may submit short or long papers, position statements, or system/demo descriptions; we will also circulate a short interest form to collect backgrounds, research interests, preferred discussion topics, demo/poster plans, and accessibility needs.

9:00–9:20 Welcome and introductions. Accessibility notes, goals, and participant introductions.

9:20–9:50 Keynote. Assistive embodied AI, accessibility, and intelligent user interfaces (20 min + 10 min Q&A).

9:50–11:00 Paper Session I. Up to 7 accepted contributions.

11:00–12:00 Demo/poster session. Prototypes, system sketches, and demonstrations.

12:00–1:15 Lunch.

1:15–2:15 Paper Session II. Up to 6 additional contributions.

2:15–3:00 Panel. Agency, autonomy, safety, shared control, personalization, evaluation, and real-world deployment.

3:00–4:00 Thematic breakouts. Small-group discussions on core workshop themes and open research challenges.

4:00–4:40 Collaboration pitches. Key ideas, tensions, research questions, and project opportunities.

4:40–5:00 Closing. Next steps and community-building.

References

- [1] David A. Abbink, Tom Carlson, Mark Mulder, Joost C. F. de Winter, Farzad Aminravan, Tricia L. Gibo, and Erwin R. Boer. 2018. A Topology of Shared Control Systems—Finding Common Ground in Diversity. *IEEE Transactions on Human-Machine Systems* 48, 5 (Oct. 2018), 509–525. doi:10.1109/thms.2018.2791570
- [2] Stephanie Arévalo Arboleda, Franziska Rucker, Tim Dierks, and Jens Gerken. 2021. Assisting Manipulation and Grasping in Robot Teleoperation with Augmented Reality Visual Cues. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, 14 pages. doi:10.1145/3411764.3445398
- [3] Ruei-Che Chang, Xirui Jiang, Rosiana Natalie, Hao Chen, Vlad Roznyatovskiy, Jianzhong Zhang, Kang G Shin, Ke Sun, and Anhong Guo. 2026. StateScribe: Towards Accessible Change Awareness Across Real-World Revisits. *arXiv preprint arXiv:2604.23749* (2026).
- [4] Ruei-Che Chang, Yuxuan Liu, and Anhong Guo. 2024. Worldscribe: Towards context-aware live visual descriptions. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–18.
- [5] Ruei-Che Chang, Rosiana Natalie, Wenqian Xu, Jovan Zheng Feng Yap, Tiange Luo, Venkatesh Potluri, and Anhong Guo. 2026. TouchScribe: Augmenting Non-Visual Hand-Object Interactions with Automated Live Visual Descriptions. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [6] Cheng-Shiu Chung, Hongwu Wang, and Rory A. Cooper. 2013. Functional Assessment and Performance Evaluation for Assistive Robotic Manipulators: Literature Review. *The Journal of Spinal Cord Medicine* 36, 4 (2013), 273–289. doi:10.1179/2045772313Y.0000000132
- [7] Jiahui Ding, Donghui Zhao, Baixue Liu, Junyou Yang, Shuoyu Wang, and Houde Lu. 2026. A safe auxiliary feeding approach based on multimodal intention recognition and flexible interaction. *International Journal of Intelligent Robotics and Applications* 10, 1 (2026), 360–384.
- [8] Hamza El Alaoui, Zakaria El Aouene, and Violetta Cavalli-Sforza. 2023. Building Intelligent Chatbots: Tools, Technologies, and Approaches. In *Proceedings of the International Conference on Innovative Research in Applied Science, Engineering, Technology* (Casablanca, Morocco). IEEE, 1–12. doi:10.1109/IRASET57153.2023.10153005
- [9] Hamza El Alaoui, Atieh Taheri, Yi-Hao Peng, and Jeffrey P Bigham. 2025. StepWrite: Adaptive Planning for Speech-Driven Text Generation. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (Busan, Republic of Korea) (UIST '25). ACM, New York, NY, USA, Article 86, 30 pages. doi:10.1145/3746059.3747610
- [10] Badr Ben Elallid, Hamza El Alaoui, and Nabil Benamar. 2023. Deep Reinforcement Learning for Autonomous Vehicle Intersection Navigation. In *Proceedings of the Conference on Innovation and Intelligence for Informatics, Computing, Technologies* (Sakheer, Bahrain). IEEE, 308–313. doi:10.1109/3ICIT60104.2023.10391453
- [11] F. Flemisch, D. A. Abbink, M. Itoh, M.-P. Pacaux-Lemoine, and G. Weßel. 2019. Joining the blunt and the pointy end of the spear: towards a common framework of joint action, human-machine cooperation, cooperative guidance and control, shared, traded and supervisory control. *Cognition, Technology & Work* 21, 4 (Aug. 2019), 555–568. doi:10.1007/s10111-019-00576-1
- [12] David Gallenberger, Tapomayukh Bhattacharjee, Youngsun Kim, and Sidhartha S. Srinivasa. 2019. Transfer Depends on Acquisition: Analyzing Manipulation Strategies for Robotic Feeding. In *Proceedings of the 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI '19)*. IEEE, 267–276. doi:10.1109/HRI.2019.8673309
- [13] Pratyusha Ghosh, Belen Liedo, and Laurel D Riek. 2026. Disability justice in human-robot interaction: Reflections on paternalism, autonomy, and care for more equitable futures. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. 754–764.
- [14] Deepak Gopinath, Siddharth Jain, and Brenna D. Argall. 2017. Human-in-the-Loop Optimization of Shared Autonomy in Assistive Robotics. *IEEE Robotics and Automation Letters* 2, 1 (2017), 247–254. doi:10.1109/LRA.2016.2593928
- [15] Anhong Guo, Xiang'Anthony' Chen, Haoran Qi, Samuel White, Suman Ghosh, Chieko Asakawa, and Jeffrey P Bigham. 2016. Vizlens: A robust and interactive screen reader for interfaces in the real world. In *Proceedings of the 29th annual symposium on user interface software and technology*. 651–664.
- [16] Guy Hoffman, Oren Zuckerman, Gilad Hirschberger, Michal Luria, and Tal Shani Sherman. 2015. Design and Evaluation of a Peripheral Robotic Conversation Companion. In *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction* (Portland, Oregon, USA) (HRI '15). Association for Computing Machinery, New York, NY, USA, 3–10. doi:10.1145/2696454.2696495
- [17] Megan Hofmann, Devva Kasnitz, Jennifer Mankoff, and Cynthia L Bennett. 2020. Living disability theory: Reflections on access, research, and design. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*. 1–13.
- [18] Siddharth Jain, Ali Farshchiansadegh, Alexander Broad, Farnaz Abdollahi, Ferdinando Mussa-Ivaldi, and Brenna Argall. 2015. Assistive Robotic Manipulation through Shared Autonomy and a Body-Machine Interface. In *2015 IEEE International Conference on Rehabilitation Robotics (ICORR '15)*. IEEE, 526–531. doi:10.1109/ICORR.2015.7281253
- [19] Yuzhi Lai, Shenghai Yuan, Peizheng Li, Boya Zhang, Benjamin Kiefer, Tianchen Deng, and Andreas Zell. 2026. FAM-HRI: Foundation-Model Assisted Multimodal Human-Robot Interaction Combining Gaze and Speech. *IEEE Transactions on Automation Science and Engineering* 23 (2026), 10521–10534. doi:10.1109/TASE.2026.3695438
- [20] Chen Liang, Yasha Irvantchi, Thomas Krolkowski, Ruijie Geng, Alanson P Sample, and Anhong Guo. 2023. BrushLens: hardware interaction proxies for accessible touchscreen interface actuation. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–17.
- [21] Chen Liang, Yuxuan Liu, Martez Mott, and Anhong Guo. 2025. HandProxy: Expanding the Affordances of Speech Interfaces in Immersive Environments with a Virtual Proxy Hand. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 9, 3 (2025), 1–30.
- [22] Dylan P. Losey, Hong Jun Jeon, Mengxi Li, Krishnan Srinivasan, Ajay Mandlekar, Animesh Garg, Jeannette Bohg, and Dorsa Sadigh. 2022. Learning Latent Actions to Control Assistive Robots. *Autonomous Robots* 46, 1 (2022), 115–147. doi:10.1007/s10514-021-10005-w
- [23] Kelly Mack, Emma McDonnell, Venkatesh Potluri, Maggie Xu, Jaily Zabalá, Jeffrey Bigham, Jennifer Mankoff, and Cynthia Bennett. 2022. Anticipate and adjust: Cultivating access in human-centered methods. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [24] Jennifer Mankoff, Gillian R Hayes, and Devva Kasnitz. 2010. Disability studies as a source of critical inquiry for the field of assistive technology. In *Proceedings of the 12th international ACM SIGACCESS conference on Computers and accessibility*. 3–10.
- [25] Harish Ram Nambiappan, Sneha Acharya, and Fillia Makedon. 2024. Multimodal Smartphone based IoT Framework for Assisting People with Disabilities in SLAM based Human Robot Interaction. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking* (Washington D.C., DC, USA) (ACM MobiCom '24). Association for Computing Machinery, New York, NY, USA, 1680–1681. doi:10.1145/3636534.3697458
- [26] Harish Ram Nambiappan, Enamul Karim, Jilur Rahman Saurav, Anushka Srivastav, and Fillia Makedon. 2022. Edge-IoT framework for speech and mobile-based human-robot interaction. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services* (Portland, Oregon) (MobiSys '22). Association for Computing Machinery, New York, NY, USA, 527–528. doi:10.1145/3498361.3538767
- [27] Harish Ram Nambiappan and Fillia Makedon. 2024. Development and Evaluation of Collision Avoidance User Interface for Assistive Vision Impaired Navigation. In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) (UIST Adjunct '24). Association for Computing Machinery, New York, NY, USA, Article 105, 3 pages. doi:10.1145/3672539.3686354
- [28] Harish Ram Nambiappan, Sama Nikanfar, Ayon Roy, Joey Hussain, Deep Shinglot, Sneha Acharya, Nicholas Gans, and Fillia Makedon. 2024. Human-Robot Interactive System for Warehouses using Speech SLAM and Deep Learning-based Barcode Recognition. In *Proceedings of the 17th International Conference on Pervasive Technologies Related to Assistive Environments* (Crete, Greece) (PE-TRA '24). Association for Computing Machinery, New York, NY, USA, 38–44. doi:10.1145/3652037.3652061
- [29] Amal Nanavati, Vinitha Ranganeni, and Maya Cakmak. 2023. Physically assistive robots: A systematic review of mobile and manipulator robots that physically assist people with disabilities. *Annual Review of Control, Robotics, and Autonomous Systems* 7 (2023).
- [30] Temitayo Olugbade, Liang He, Perla Maiolino, Dirk Heylen, and Nadia Bianchi-Berthouze. 2023. Touch Technology in Affective Human-, Robot-, and Virtual-Human Interactions: A Survey. *Proc. IEEE* 111, 10 (2023), 1333–1354. doi:10.1109/JPROC.2023.3272780
- [31] Akhil Padmanabha, Janavi Gupta, Chen Chen, Jehan Yang, Vy Nguyen, Douglas J Weber, Carmel Majidi, and Zackory Erickson. 2024. Independence in the home: A wearable interface for a person with quadriplegia to teleoperate a mobile manipulator. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 542–551.
- [32] Akhil Padmanabha, Jessie Yuan, Janavi Gupta, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, and Zackory Erickson. 2024. Voicepilot: Harnessing LLMs as speech interfaces for physically assistive robots. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–18.
- [33] Max Pascher, Annalies Baumeister, Stefan Schneegass, Barbara Klein, and Jens Gerken. 2021. Recommendations for the Development of a Robotic Drinking and Eating Aid: An Ethnographic Study. In *Human-Computer Interaction – INTERACT 2021*. Springer, 331–351. doi:10.1007/978-3-030-85623-6_21
- [34] Max Pascher, Franz Felix Goldau, Kirill Kronhardt, Udo Frese, and Jens Gerken. 2024. AdaptIX: A Transitional XR Framework for Development and Evaluation of Shared Control Applications in Assistive Robotics. *Proceedings of the ACM on Human-Computer Interaction* 8 (2024). Issue EICS. doi:10.1145/3660243

- [35] Max Pascher, Uwe Gruenefeld, Stefan Schneegass, and Jens Gerken. 2023. How to Communicate Robot Motion Intent: A Scoping Review. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery. doi:10.1145/3544548.3580857
- [36] Max Pascher, Kirill Kronhardt, Til Franzen, and Jens Gerken. 2022. Adaptive DoF: Concepts to Visualize AI-Generated Movements in Human-Robot Collaboration. In *Proceedings of the 2022 International Conference on Advanced Visual Interfaces*. ACM. doi:10.1145/3531073.3534479
- [37] Max Pascher, Kirill Kronhardt, Til Franzen, Uwe Gruenefeld, Stefan Schneegass, and Jens Gerken. 2022. My Caregiver the Cobot: Comparing Visualization Techniques to Effectively Communicate Cobot Perception to People with Physical Impairments. *Sensors* 22, 3 (2022). doi:10.3390/s22030755
- [38] Max Pascher, Kirill Kronhardt, Franz Felix Goldau, Udo Frese, and Jens Gerken. 2023. In Time and Space: Towards Usable Adaptive Control for Assistive Robotic Arms. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication*. IEEE, 2300–2307. doi:10.1109/RO-MAN57019.2023.10309381
- [39] Max Pascher, Kevin Zinta, and Jens Gerken. 2024. Exploring Discrete and Continuous Input Control for AI-Enhanced Assistive Robotic Arms. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. ACM. doi:10.1145/3610978.3640626
- [40] Vinitha Ranganeni, Varad Dhat, Noah Ponto, and Maya Cakmak. 2024. Accessteleopkit: A toolkit for creating accessible web-based interfaces for tele-operating an assistive robot. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–12.
- [41] Vinitha Ranganeni, Noah Ponto, and Maya Cakmak. 2023. Evaluating customization of remote tele-operation interfaces for assistive robots. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 1633–1640.
- [42] Fraser Robinson and Goldie Nejat. 2022. An analysis of design recommendations for socially assistive robot helpers for effective human-robot interactions in senior care. *Journal of Rehabilitation and Assistive Technologies Engineering* 9 (2022), 20556683221101389.
- [43] Ofir Sadka, Hadas Erel, Andrey Grishko, and Oren Zuckerman. 2018. Tangible interaction in parent-child collaboration: encouraging awareness and reflection. In *Proceedings of the 17th ACM Conference on Interaction Design and Children (Trondheim, Norway) (IDC '18)*. Association for Computing Machinery, New York, NY, USA, 157–169. doi:10.1145/3202185.3202746
- [44] Sushil Kumar Sahoo and Bibhuti Bhusan Choudhury. 2023. A review on human-robot interaction and user experience in smart robotic wheelchairs. *Journal of Technology Innovations and Energy* 2, 2 (2023), 39–55.
- [45] Atieh Taheri, Arthur Caetano, and Misha Sra. 2024. Virtual steps: The experience of walking for a lifelong wheelchair user in virtual reality. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 168–178.
- [46] Atieh Taheri, Hamza El Alaoui, Patrick Carrington, and Jeffrey P Bigham. 2026. “I followed what felt right, not what I was told”: Autonomy, Coaching, and Recognizing Bias Through AI-Mediated Dialogue. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 1–23.
- [47] Atieh Taheri, Carlos Gilberto Gomez-Monroy, Vicente Borja, and Misha Sra. 2024. Mouseclicker: Exploring tactile feedback and physical agency for people with hand motor impairments. *ACM Transactions on Accessible Computing* 17, 1 (2024), 1–31.
- [48] Atieh Taheri, Misha Sra, Patrick Carrington, and Jeffrey P Bigham. 2025. Designing Through Lived Experience: Reflections on Control, Embodiment, and Social Bias in Accessibility Research. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–7.
- [49] Atieh Taheri, Ziv Weissman, and Misha Sra. 2021. Design and evaluation of a hands-free video game controller for individuals with motor impairments. *Frontiers in Computer Science* 3 (2021), 751455.
- [50] Atieh Taheri, Ziv Weissman, and Misha Sra. 2021. Exploratory design of a hands-free video game controller for a quadriplegic individual. In *Proceedings of the Augmented Humans International Conference 2021*. 131–140.
- [51] Stephanie Valencia, Richard Cave, Krystal Kallarackal, Katie Seaver, Michael Terry, and Shaun K Kane. 2023. “The less I type, the better”: How AI language models can enhance or impede communication for AAC users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [52] Stephanie Valencia, Michal Luria, Amy Pavel, Jeffrey P. Bigham, and Henny Admoni. 2021. Co-designing Socially Assistive Sidekicks for Motion-based AAC. In *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction (Boulder, CO, USA) (HRI '21)*. Association for Computing Machinery, New York, NY, USA, 24–33. doi:10.1145/3434073.3444646
- [53] Michael Walker, Hooman Hedayati, Jennifer Lee, and Daniel Szafr. 2018. Communicating Robot Motion Intent with Augmented Reality. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction (HRI '18)*. Association for Computing Machinery, 316–324. doi:10.1145/3171221.3171253
- [54] Michael E. Walker, Thao Phung, Tathagata Chakraborti, Tom Williams, and Daniel Szafr. 2023. Virtual, Augmented, and Mixed Reality for Human-Robot Interaction: A Survey and Virtual Design Element Taxonomy. *ACM Transactions on Human-Robot Interaction* 12, 4, Article 43 (2023), 39 pages. doi:10.1145/3597623
- [55] Jessie Yuan, Janavi Gupta, Akhil Padmanabha, Zulekha Karachiwalla, Carmel Majidi, Henny Admoni, and Zackory Erickson. 2024. Towards an llm-based speech interface for robot-assisted feeding. In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–4.

A Organizers

Hamza El Alaoui is a Ph.D. student at CMU's School of Computer Science. His research focuses on cyber-physical human augmentation, combining machine learning, embedded systems, robotics, and HCI to create systems that extend human communication, perception, and physical capabilities. His work spans language technologies, multimodal sensing, human-AI integration, and accessibility systems. He previously worked on machine learning at Oracle and was a visiting scientist at Sony Computer Science Laboratories.

Harish Ram Nambiappan received his Ph.D. in Computer Science from The University of Texas at Arlington. His research focuses on developing assistive navigation and robotic systems that enable BLV people to navigate dynamic indoor environments and interact directly with robots. He has built cross-modal interfaces that allow users to locate objects, direct robotic manipulation, and remain safe around moving robots and people. His broader work also includes autonomous service robots for healthcare, placing his research at the intersection of robotics, accessibility, autonomy, and HRI.

Atieh Taheri is a Presidential Postdoctoral Fellow at CMU. Her research combines accessibility, human-centered AI, mixed reality, and participatory design to build interactive systems that expand agency and creative expression for people with disabilities. Her work includes hands-free controllers, tactile interfaces, accessible conversational systems, VR experiences, and accessible generative-AI tools. She previously conducted research at Google and Apple, and earned her Ph.D. in Electrical & Computer Engineering from the University of California, Santa Barbara.

Chen Liang is a Ph.D. candidate in Computer Science & Engineering at the University of Michigan. His research develops human-AI systems that make UIs more intelligent, adaptable, and accessible. He creates AI-powered interaction proxies that translate or automate actions on users' behalf without requiring modifications to the underlying interface. He previously conducted research at Adobe.

Max Pascher is a Postdoctoral Researcher at TU Dortmund University. His research focuses on AI-enhanced human-robot collaboration, including shared-control techniques for assistive robotic arms, intervention strategies for autonomous robotic tasks, and XR-based tools for designing and evaluating human-robot interfaces. He also works on multimodal interaction and methods that keep users in control of autonomous robotic systems. He received his Ph.D. in Human-Robot Interaction from the University of Duisburg-Essen.

Vinitha Ranganeni is the Human-Robot Interaction Lead at Hello Robot, where she leads the development and real-world evaluation of human-centered software for the Stretch mobile manipulator. Her work focuses on accessible assistive robotics for people with disabilities, including customizable interfaces and in-home deployments that support greater independence. Her work has been featured in *IEEE Spectrum* and the *IEEE Spectrum Technology*. She received her Ph.D. in Computer Science from the University of Washington.

Stephanie Valencia is an Assistant Professor of Information Science at the University of Maryland. Her research focuses on equitable access to assistive technology, open-source hardware, and the design of communication systems that increase user agency, and accessibility. Using participatory design, she develops AI-enabled and embodied expressive technologies for AAC users, with an emphasis on building, deploying, and rigorously evaluating systems in real-world settings. She received her Ph.D. in HCI from CMU.

Zackory Erickson is an Assistant Professor at CMU's Robotics Institute and director of the RCHI Lab. His research develops robot-learning, mobile-manipulation, sensing, and control methods for physical HRI to help older adults and people with disabilities live independently. His work spans assistance with feeding, dressing, bathing, and personal care; wearable interfaces for limited hand function; simulation; haptic perception; and safe, personalized robot autonomy. He received his Ph.D. in Robotics from Georgia Tech.

Tapomayukh "Tapo" Bhattacharjee is an Assistant Professor of Computer Science at Cornell University, where he directs the EmPRISE Lab. His research focuses on physical robot caregiving and physical HRI, particularly assistive robots that support people with mobility limitations in activities of daily living. His lab takes a full-stack robotics approach spanning sensing, perception, planning, learning, control, manipulation, haptic perception, and tactile sensing. He received his Ph.D. in Robotics from Georgia Tech and was a Postdoctoral Researcher at the University of Washington.

Anhong Guo is an Assistant Professor of Computer Science at the University of Michigan. His research lies at the intersection of HCI, AI, and accessibility. He develops personal assistive technologies that adapt to users' individual abilities, contexts, and intentions, particularly for BLV people. His work spans human-AI systems for accessing visual information, accessible physical and touchscreen interfaces, customizable visual-assistance tools, context-aware real-world descriptions, augmented and virtual reality, and multimodal end-user programming. He received his Ph.D. in HCI from CMU.

Eshed Ohn-Bar is an Assistant Professor of Electrical & Computer Engineering at Boston University and leads the Human-to-Everything Lab. His research develops machine-intelligence systems for embodied, assistive, and autonomous applications, spanning perception, learning, planning, control, and human-machine interaction. His work includes accessible navigation and AI agents that provide real-time assistance to BLV people. He received his Ph.D. in Electrical Engineering from the University of California, San Diego, and was a Humboldt Fellow at the Max Planck Institute.

Jeffrey P. Bigham is an Associate Professor at CMU's Human-Computer Interaction Institute and Language Technologies Institute, where he directs the Beneficial Intelligence Group. He also serves as Director of Human-Centered Machine Intelligence & Responsible AI at Apple, leading research on AI/ML design, accessibility, and UI understanding. His research combines machine learning and human computation to develop reliable, human-centered AI systems for real-world use, particularly in accessibility. He received his Ph.D. in Computer Science from the University of Washington.

B Intended Audience and Outcomes

The intended audience includes researchers, practitioners, designers, and students working across UIST, accessibility, human–robot interaction, robotics, assistive technology, rehabilitation, AI, AR/VR, and interactive systems. We especially welcome participants interested in how people with physical, sensory, communication, and cognitive impairments interact with emerging assistive embodied AI systems, including researchers and designers with lived experience of disability.

Expected outcomes include:

- (1) A shared research agenda for UIST-relevant work at the intersection of assistive embodied AI systems, accessibility, and human-centered interaction.

- (2) A set of concrete design challenges and use-case scenarios spanning physical assistance, caregiving, teleoperation, shared autonomy, multimodal control, visual access, navigation, and real-world deployment.
- (3) New collaborations across the UIST, accessibility, HRI, robotics, rehabilitation, AI, and assistive technology communities.
- (4) A public workshop website at embodiedaccess.github.io hosting workshop information, accepted contributions, demos, participant interests, resources, and discussion outcomes.